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Project Title:	Granular Flow Instabilities: Transients, Aging, and Segregation Dynamics		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
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Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA GRC
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The aim of this project is an interdisciplinary study of granular flow instabilities, involving both laboratory experiments for the development of physical models, as well as approaches and data sets from geology. The central aim of the proposed work is the improvement of physical models for granular flow instabilities, and the application of those models to rock avalanches.
The proposed research will provide experimental data and test theoretical models to address open questions about three main aspects of granular flow instabilities:

- Flow initiation and cessation: Granular matter can exhibit a sudden transition from a solid-like state to a flowing state and vice versa, e.g. during avalanching and jamming. We will investigate particle motions during such transients. One

Task Description:

question is, what timescales and spatial scales best describe such a transition?

- Dynamics of aging: Even an apparently stationary granular assembly slowly rearranges (aging). This causes changes in material properties, such as the gradual increase in yield strength. We will measure, the changes in the network of particle contacts directly with confocal imaging.

- Dynamics of segregation: Grains of different size or shape will segregate under some conditions in a driven granular material. While models for segregation exist, several fundamental issues warrant further study. We will relate segregation to the sharing of momentum and energy between particle species.

Understanding granular flow instabilities is of fundamental importance since it may reveal the most suitable equations to describe granular flows. Flow instabilities are also of great technological significance for a broad range of industries. The sudden jamming of grains in confined geometries, e.g. in a hopper, is a significant problem for food, mining, and raw materials processing industries. The slow creep and gradual strengthening of a quasi-stationary granular pile under stress can lead to structural damage of storage containers. The gradual segregation of poly-disperse mixtures of granular matter during shear is especially dangerous in the pharmaceutical industries, where active ingredients for drugs need to be uniformly mixed in powder form.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Two main impacts are in improved understanding of rock avalanche hazards. The results were communicated through two paper in the geology literature:
1) The basic physics of long runout avalanches. One series of experiments has been aimed at recreating long runout rock avalanches in the laboratory. Such avalanches are extremely rare, but highly destructive events where a large part of a mountain slides very far into a valley at high speed. The distances traveled by such long runout avalanches far exceed what one might expect due to friction between rocks. Modeling such events in the lab indicated that several simple flow geometries cannot recreate long runout avalanches, a result reported in the International Journal of Earth Sciences.

2) By measuring how the start of granular flow depends on the direction of prior shear, we indirectly found preferred directions in the particle contact network which influence jamming and flow (Toiya, Stambaugh et al. Physical Review Letters 2004). In an application of this insight, we demonstrated that direction of prior shear may be a useful parameter in assessments of rock avalanche hazards (Losert et al, J Geophysical Research 2003).

Task Progress:

This project resulted in 9 publications, plus 4 papers in preparation. It was also the focus of many invited talks, colloquia and seminars as listed below. Two students working on this project are graduating in the summer of 2006. In addition, one visiting student from the University of Rennes was supported by this grant, and several undergraduate students contributed to this effort. The students won several awards for their research. Our main results are described below. Beyond the work on long runout avalanches highlighted in the impact section, the project focused on two tasks:

1) Understanding the onset of flow under shear or penetrometer forcing. This work included the design of a novel 3D imaging technique to analyze particle arrangements in three dimensions. 2) Analyzing the flow of mixtures of particles, in particular determining the mechanism for segregation in a flow of mixtures. We developed a new model based on the differences in native flow speed and shear band width that agrees with both microscopic and macroscopic observations.

Segregation of mixtures:

We studied granular particles, like sand or glass beads, that are mixed in a partially filled, horizontal, cylindrical drum. When the drum is rotated, it is observed that there is a flowing layer of grains on the free surface of the granular medium. In addition, if the particles have different sizes, spatial segregation of the particles by size is observed. This segregation occurs in two phases. During the first phase, called radial segregation, the smaller particles form a radial core. In the second, called axial segregation, particles segregate into alternating bands along the axis of the drum.

We performed a detailed study of the characteristics of the flow to determine the physical mechanisms driving axial segregation: We characterized the top surface of the flowing layer by tracking particles using a high speed camera. We then extracted average quantities such as velocity and diffusion. The average velocities show surprising behavior: Particles in small particle bands have a higher downhill flow velocity than particles in large particle bands. We also observed that there is a pattern of sideways velocity as a function of position down the flow. Particles flow into small particle bands in the middle of the flow but flow out of small particle bands at the bottom.

We presented the framework for a new model based on our experimentally observed results. We explain the axial band formation in terms of the observed surface flow patterns. We show how two physical processes could contribute to the band formation: A.) Accelerating granular material does not necessarily collide while decelerating granular material requires collisions. B.) Different size particles flow at different velocities. Our framework connects differences in flow velocities on the surface of the drum with the radial segregation in the bulk of the drum.

We compared these results to current models, including models by Savage, Zik, Aranson, and Elperin. We tested the general model assumption that the particles always flow in the direction of steepest descent by measuring the surface height of the banded state with a laser line. We found that although there is some indication that particles flow in the direction of steepest descent, there is strong flow that is not in the direction of steepest descent, contrary to the given models.

Finally, we study oscillating patterns in mixtures of three sizes of glass beads. Ternary mixtures of particles form bands within bands of the different particle sizes. For certain experimental conditions we observe traveling and oscillating patterns. We analyze these patterns in light of the different models for granular band formation.

Onset of granular flows and memory effects under penetrometer and shear forcing:

Shear flows are induced in a shear cell of Taylor-Couette type by moving a boundary wall. We find that how a granular shear flow starts depends strongly on the prior shear direction. We observe that when the shear direction is reversed, the material goes through a transient period during which the material compacts, the shear force is small, and the shear band is wide. Three dimensional confocal imaging of particle rearrangements during shear reversal shows that bulk and

surface flows are comparable.

Local flows are induced by forcing a rod into a fluid immersed granular bed with various preparation methods. Particle rearrangements are observed in 3D by confocal microscopy and by moving a laser sheet through the sample. Image analysis indicates that rearrangements spread farthest not directly under the penetrometer but in a ring around the penetrometer. In addition, the direction of preformed stress chains in the material influences the particle rearrangements. Material compressed from one side exhibits anisotropic particle rearrangements under penetrometer testing.

Particle rearrangements that do not lead to steady flows but overall increase in the packing fraction have also been investigated. Vibration-induced compaction of spherical grains indicates that in addition to the cage motion there are rare ”jump” events in which a single particle moves significantly more compared to its neighbors. Although rare, such ”jumps” play a significant role in the compaction of the material as a whole. Temperature cycling experiments show similar results. Compaction of material has also been observed in the the Couette cell. Repeated reversals or oscillations of the shear direction lead to additional compaction, which can be described by a stretched exponential, similar to compaction induced by tapping.

For 3D imaging of dry and non-transparent materials, we used X-ray microtomography in collabroration with the Univ. of Rennes. Images taken during the ongoing penetrometer studies of spherical particles and an image of the lunar simulant JSC-1 are shown in the pdf of the task report.

Student Training and outreach: I have carried out a range of outreach activities. The main contribution was a table of activities at Maryland Days that demonstrated the unique properties of granular materials, and how current research relates to everyday life. My table, manned by my graduate and undergraduate students includes a demo of spontaneous separation of sand by size, and a hands on demonstration of how strong jammed sand – both spherical and rod-shaped - can be. This table has been visited my more than 1000 visitors each year.

Students have won a number of awards for their work on this project including:

1. Cosmos Club Fellowship (Spring 2003: Mike Newey) - presentation at Cosmos Club foundation dinner April 2004. 2. Cosmos Club Fellowship (Spring 2003: Masahiro Toiya) - presentation at Cosmos Club foundation dinner April 2004. 3. Undergraduate Research Fellowship in IREAP (Spring 2003: Dominic Britti, Fall 2004, Andrew Porter) 4. Best Graduate Student Presentation Award, IREAP, (Fall 2003 and Spring 2006: Masahiro Toiya) 5. Undergraduate student presentation award TREND REU program (Summer 2004, Katherine Newhall, Summer 2005, Kenneth Desmond) 6. Best Poster Award– Burgers Fluid Dynamics Meeting, Univ of Maryland (November 2004, Masahiro Toiya) 7. IPST - Monroe Martin Prize for undergraduate research (May 2005, Ricardo Pizarro).

Bibliography Type:	Description: (Last Updated: 04/11/2007)
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Awards	Porter A. "Undergraduate Research Fellowship in IREAP, Fall 2004." Oct-2004

Awards	Toiya M. "Best Graduate Student Presentation, IREAP, Fall 2003." Oct-2003
Awards	Toiya M. "Best Graduate Student Presentation, IREAP, Spring 2006." Apr-2006
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